

Supplemental material for XYZ

1 SUMMA details

At CSSL, the peak water equivalent was XYZ higher in the open site relative to the forest site leading up to the heatwave event which may have resulted from a combination of snow interception or antecedent melt. We found good correspondence between SUMMA and observations in the open and forest sites, respectively. This was found after adjusting the rain/snow threshold temperature to XYZ, the canopy wind reduction coefficient to and a snow surface roughness of 0.002 m for both sites. These values are well within established bounds for those parameters (CITE). Comparisons against the forest site CNR4 radiometer showed good correspondence between the SUMMA modelled below-canopy shortwave radiation, lending confidence to the prescribed LAI value and UEB 2-stream shortwave parameterization implemented in SUMMA.

2 Cirrus and RRTMG

Then, we insert a hypothetical cirrus cloud at 200 hPa in the model with a cloud water path of 6 gmtwo and an effective cloud radii of todo typical, for mid-latitude cirrus cloud todo $cCRF_x$ is computed as the difference between the four component net radiative difference between the cloudy atmosphere simulation and the clear-sky simulation.

3 Additional details about SUMMA

The bulk expression for the H_s implemented in SUMMA is given by:

$$H_s = c_p \rho C_s u (T_{air} - T_{snow}), \quad (1)$$

where c_p is the specific heat capacity of dry air (constant), u is the horizontal wind speed, ρ is the dry air density, and C_s is the bulk transport coefficient. In this work, we use the SUMMA formulation for C_s , which models the stability correction as a function of the bulk Richardson number.

We use the “Loisinv” stability correction function implemented in the SUMMA model and assume that $z_h = z_0$

$$C_h = \frac{k^2}{\ln(z/z_0)^2} \cdot f(Ri) \quad (2)$$

Where Ri is the bulk Richardson number, defined as

$$Ri = \frac{9.81(T_{air} - T_{snow})}{0.5(T_{air} + T_{snow})u^2}. \quad (3)$$

and $f(Ri)$ is a function with range (0,1] that declines as a function of increasing Ri.

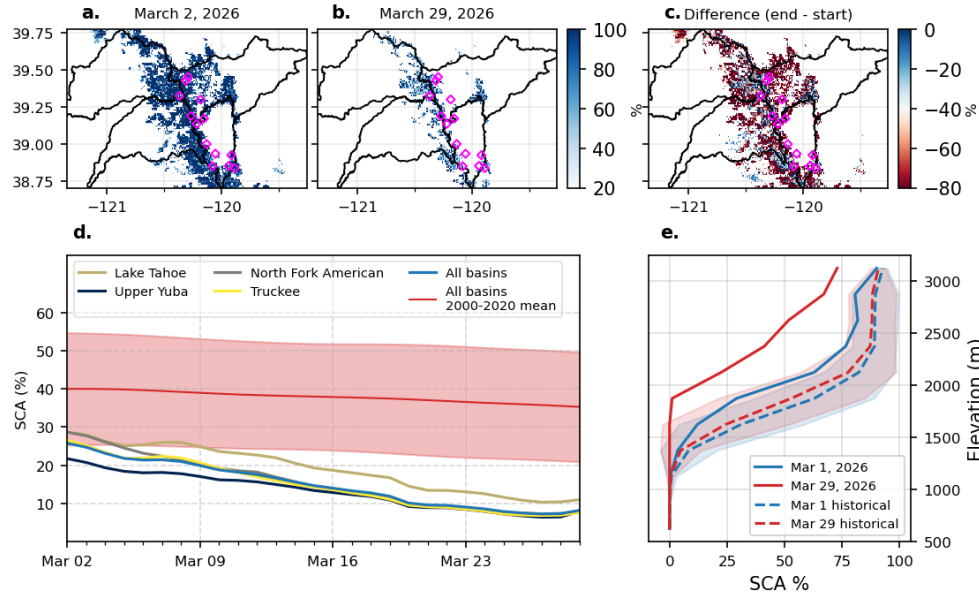


Figure S1: SPIRES SCA across the study region during the March heatwave. Watershed boundaries shown in black and SNOTEL monitoring sites indicated by magenta diamonds. **(a, b, c)** SCA (%) on March 2 and March 29, 2026, respectively, and the difference between the two dates. **(d)** Daily mean SCA (%) for each individual watershed and aggregated across all four basins. The 2000–2020 historical mean and ± 1 standard deviation are depicted. **(e)** Mean SCA as a function of elevation band (250 m bins) for March 2 (blue) and March 29 (red), shown alongside historical values for the same dates (dashed lines) with ± 1 standard deviation shading.

4 Validation

As a measure of the canopy temperature, we use data from the VIIRS/Suomi-NPP VNP21A1D VIIRS Version 1 Land Surface Temperature (LST) data product for day and nighttime. The canopy temperature data has a resolution of 1km and provides twice-daily observations of LST.

4.1 Idealized scaling of energy fluxes with T_{air} and RH over melting snow

The results in Sec. purpleTODO: SECTION — and the relatively small contributions of H_l and H_s despite anomalously high temperatures — can best be explained by considering “offline” test cases assuming a fixed 0 black°C melting snow surface. Several important considerations arise since snow cannot exceed this thermodynamic limit. In these conditions, turbulent exchange between the snow and an above 0 black°C atmosphere will always be limited by the effects of buoyant suppression Brutsaert2013-ef.

Case in point, H_s over melting 0 black°C snow increases linearly as a function of black T_{air} with a slope determined by black w_{spd} , assuming an un-physical neutral condition without taking into account the effects of the buoyant suppression of turbulence black(Fig. ??a). However, accounting for buoyant suppression decreases H_s relative to the neutral assumption, so much so that for moderate black w_{spd} values of 5 black $\text{m}\cdot\text{s}^{-1}$, H_s actually *decreases* as a function of increasing black T_{air} above 4black°C. To better interrogate this relationship, black(Fig. ??c) depicts the black T_{air} value that maximized H_s for a given black w_{spd} . A black w_{spd} of 5 black $\text{m}\cdot\text{s}^{-1}$ has a corresponding H_s maximum at 4-5 black°C; the stability effect will reduce H_s at black T_{air} values above or below this black(Fig. ??c), and a black w_{spd} of 2 black $\text{m}\cdot\text{s}^{-1}$ achieves a maximum H_s when black T_{air} is only a degree or two above freezing.

Evaporation and/or sublimation is likewise impacted by the 0 black°C limitation — to a reasonable approximation, the vapor pressure of pooling liquid water or 0black°C C ice crystals on top of the snow never exceeds 6.11 hPa. Therefore, the vapor pressure of the air must be less than 6.11 hPa for sublimation or evaporation to occur. Assuming a constant RH of 20% (roughly the minimum observed during the March heatwave), H_l decreases rather than increases as a function of black T_{air} over a 0black°C snow regardless of black w_{spd} black(Fig. ??b,d). black T_{air} greater than approximately 24-25black°C at 20% RH *adds energy to the snow* via condensation/deposition. The effects of thermodynamic stability further reduce H_l as a function of black T_{air} . In reality, though, RH is not independent of black T_{air} , and generally decreases as black T_{air} rises during the daytime due to water vapor conservation and dry-air entrainment from the upper levels of the boundary layer purpleTODO: CITE. Sublimation is permitted across the full range of RH values as black T_{air} approaches 0black°C. At 25black°C (nearly the maximum observed during the March heatwave), RH values must be less than 20% for sublimation to occur.

Assuming an emissivity value of 0.98 Huang2016-vq snow emits outgoing black $LW \uparrow$ at a maximum rate of 309 black $\text{W}\cdot\text{m}^{-2}$. The effective emissivity of the clear-sky atmosphere, as well as black $LW \downarrow$, can be easily visualized by considering the [?] formula which is an approximation of the full radiative transfer equations implemented in RRTMG and other models (Sec ??). In addition to scaling to the fourth power, small variations in RH have

a large impact on $\text{black}LW \downarrow$. Over melting snow, a $\text{black}T_{\text{air}}$ of 20°C and a low RH of 20% is unable to overcome the counteracting emission by the snowpack, and $\text{black}LW_{\text{net}}$ is negative on the order of $20 \text{ blackW}\cdot\text{m}^{-2}$. For the same $\text{black}T_{\text{air}}$, an RH of 60% more than doubles the magnitude of $\text{black}LW_{\text{net}}$ and yields a net-heating of the snow. Based on these considerations, snow is highly sensitive to variations in both temperature and also atmospheric humidity.

Taken together, increasing RH and $\text{black}T_{\text{air}}$ will tend to increase $\text{black}LW_{\text{net}}$ and also *decrease* cooling via the sublimation/evaporation mechanism. While $\text{black}LW_{\text{net}}$ may not always be a heat-source, but rather a heat-sink, mechanism for snowmelt rates, $\text{black}LW_{\text{net}}$ over melting snow increases non-linearly and monotonically as a function of rising $\text{black}T_{\text{air}}$ and is completely insensitive to wind motions $\text{black}LW_{\text{net}}$ is perhaps best thought of as “handbrake” on the rate of snowmelt; rising $\text{black}T_{\text{air}}$ during heatwaves means that this brake is applied less and less. The same is not true for H_s which only scales linearly with $\text{black}T_{\text{air}}$ at very high $\text{black}w_{\text{spd}}$ values due to the counteracting effects of buoyancy.

We caution that this is only one possible parameterization for describing the effects of surface layer stability on turbulent exchange, and the assumptions of the theory (while employed in virtually all land, snow, and atmospheric models), may not hold for complex mountain terrain. Terrain mesoscale circulations, in which case turbulence isn’t generated by shear between the snow surface and atmosphere, but may be generated by other features and advected over the snow, could yield greater rates of H_s or H_l for given meteorological conditions.

4.2 Vegetation impedes melt during the heatwave

While we find that $\text{RF}_{\text{Canopy},LW}$ scales linearly with rising $\text{black}T_{\text{air}}$, meaning that the vegetation LW effect is exacerbated during heatwaves (rather than constant regardless of $\text{black}T_{\text{air}}$), the net-effect ($\text{CanRE}_{\text{net}}$) transitioned from a net-positive (vegetation enhances energy entering the snow) to net-negative (vegetation reduces energy entering the snow). This finding matches the observational data; while snow ablated earlier in the forest site, the rate of melt was less than the open `purpleTODO`: how much). The earlier ablation was likely explained by canopy interception, since there was less SWE at the beginning of March in the forest site compared to the open. The ablation rate is explained by both the declining snow albedo increasing the sensitivity to $\text{black}SW \downarrow$ and rising sun-angles throughout March. In other words, the SW-shading effect of the canopy is relatively minimal when the snow is bright and the strength of the sun is weak and fails to outweigh the added LW effect.

Given that this heatwave occurred early in the season compared to those observed in `purpleTODO`: cite Rhoades, we hypothesize that same analysis of $\text{CanRE}_{\text{net}}$ would likely demonstrate a stronger cooling effect during later-season heatwaves with stronger solar insolation. Therefore, we conclude that *moderate canopy cover yields snowpacks that are more resilient to heatwaves than equivalent open areas* given that solar radiation is still such a strong snowmelt forcing during such conditions. This finding is consistent with reduced snow duration following the Caldor fire in the American river basin, immediately to the south of our study region Cowherd2026-qy. Hatchett2023 likewise found that mid-winter dryspells increase snowmelt rates in post-fire environments in the Sierra nevada due to increased solar insolation from canopy loss by wildfire.

4.3 Elevation, slope, and aspect

We only consider flat, non sloping surfaces in this analysis (with a by-definition undefined aspect). In reality the slope and terrain aspect have a substantial influence on the $\text{black}SW \downarrow$ incident on the snowpack, as well as $\text{black}LW \downarrow$ from surrounding terrain.

4.4 The role of antecedent cold-content

- The amount of energy required to overcome the cold content is far less than what is required to melt ice.

4.5 Heatwaves in the future and implications for water management

- CMIP6 models underestimate monthly high temperature extremes by 11–12% Duan2025-rl.
- Statistical WSFS models may overestimate runoff during anomalously warm periods [?] [?] if historical "runoff ratios" (the amount of streamflow relative to precipitation) are assumed in the model. We have shown that such reductions in runoff ratio are more likely to be caused by soil evapotranspiration rather than snow sublimation.
- This heatwave happened before April 1, which is the date that is historically used for water resource management decisions. Post April heatwaves may be more challenging for water management under current practices — this happened in the Spring of 2022 purpleTODO: CITE DWR Interview. Once allocations have been made by April, early snowmelt could prove costly for subsequent decisions informed by expected allocations such as planting acreage or types of crops that now may not have water available for later season irrigation. Similarly, municipal water agencies may not receive expected water late in the season. Fish, recreation, same problems? Basically, snoweaters mess everyone up later on!
- "Slower snowmelt in a warmer world" concept Musselman2017-av probably still holds but merits additional investigation. E.g., 2023 had rates on the order of 3 in/day in July compared to the 1-1.5 in/day during the March heatwave of 2026. The point is that the max rate of snowmelt for the entire water year may decline, but early season heatwaves are already increasing the melt rates for a given month or day of year.
- purpleTODO: Discuss $\text{black}\alpha$ decline mechanisms — how does liquid water impact $\text{black}\alpha$ and how is this $\text{black}T_{\text{air}}$ dependent

4.6 Data Limitations

Things here.